

Case 16: No-spread limiting behavior

1 Verification purpose

Case ID: CASE16_NSP. This unit test verifies the zero-normal-velocity limit of the ELMFIRE level-set pathway. A zero ADJ raster suppresses the base spread rate while wind, slope, spotting, crown fire, and suppression are disabled. The case exercises input multiplication, propagation, and phi/vs output and can expose ignored adjustment rasters, nonzero numerical forcing, or unintended growth. It is implementation verification, not physical validation. The question is whether the final rate remains below $10^{-6} \text{ m min}^{-1}$ and the ignited footprint remains confined to the initialization band.

2 Mathematical formulation and numerical implementation

The propagation equation is $\partial_t \phi + V_n |\nabla \phi| = 0$. The case sets the multiplicative adjustment to zero, so $V_n = 0$ and therefore $\partial_t \phi = 0$. ELMFIRE still initializes a grid-resolved ignition neighborhood, so exact zero burned area is not expected; the invariant is no subsequent outward propagation.

3 Derivation of expected behavior

Integrating $\partial_t \phi = 0$ gives $\phi(\mathbf{x}, t) = \phi(\mathbf{x}, 0)$. Thus the output rate is zero and any negative- ϕ area must be limited to the rasterized initialization band. A 2.5-cell equivalent-radius allowance bounds that band without treating subcell front placement as spread.

The generated pre-ignition ϕ raster contains no negative cells and the adjustment raster is identically zero. Thus

$$V_n = R_0(ADJ) = R_0(0) = 0, \quad \phi(\mathbf{x}, 10 \text{ s}) = \phi(\mathbf{x}, 0), \quad R_{\text{ref}} = 0 \text{ m min}^{-1}.$$

The rate metric is therefore the absolute quantity $e_R = \max_i |R_i|$, with the substituted test $e_R \leq 10^{-6} \text{ m min}^{-1}$; a relative error is undefined for the zero reference. ELMFIRE may rasterize its point ignition into a small initialization footprint, so the spatial metric is

$$r_A = \sqrt{\frac{N_b (1 \text{ m})^2}{\pi}},$$

where N_b counts final cells with $\phi < 0$. Its no-propagation reference is zero and the explicit grid allowance is $r_A \leq 2.5(1 \text{ m}) = 2.5 \text{ m}$.

4 Verification metrics and acceptance criteria

The maximum absolute finite final vs value tests the zero-velocity invariant directly and must not exceed $10^{-6} \text{ m min}^{-1}$. The equivalent burned radius $r_A = \sqrt{N_b \Delta x^2 / \pi}$ must be no more than $2.5 \Delta x$; this whole-domain area measure detects growth in any direction and is more robust than one transect. Nodata is excluded, and the last matching final phi and vs rasters are required. Both conditions must pass; missing output is NOT EVALUABLE.

5 Simulation configuration and predicted outcome

The generated EPSG:32610 domain is 200 m square on a 200-by-200 grid. Fuel, moisture, and terrain are uniform; wind and slope are zero; ADJ=0; the ignition is centered; and duration is 10 s. The predicted vs field is zero and the final ignition footprint does not expand beyond its grid initialization band.

Parameter	Configured value	Units	Role
Domain/grid	200 by 200 cells; 1	m/cell	Resolves the stationary ignition
Initial front	centered rasterized band	—	Nonzero initialized footprint
Adjustment	ADJ=0	—	Forces zero normal speed
Time control	step 1; stop 10	s	Tests the zero-speed invariant
Disabled forcing	wind, slope, crown, spotting, suppression	—	Excludes alternate growth pathways

Figure 1 is generated mechanically from the prepared case-local GeoTIFF inputs. It identifies the selected source artifact and representative variant; the left panel shows a fuel, terrain, moisture, wind, canopy, or other controlling field selected for this experiment, and the right panel shows the cells selected by the non-positive initial level-set field. No values in this figure are redrawn or inferred from the expected response.

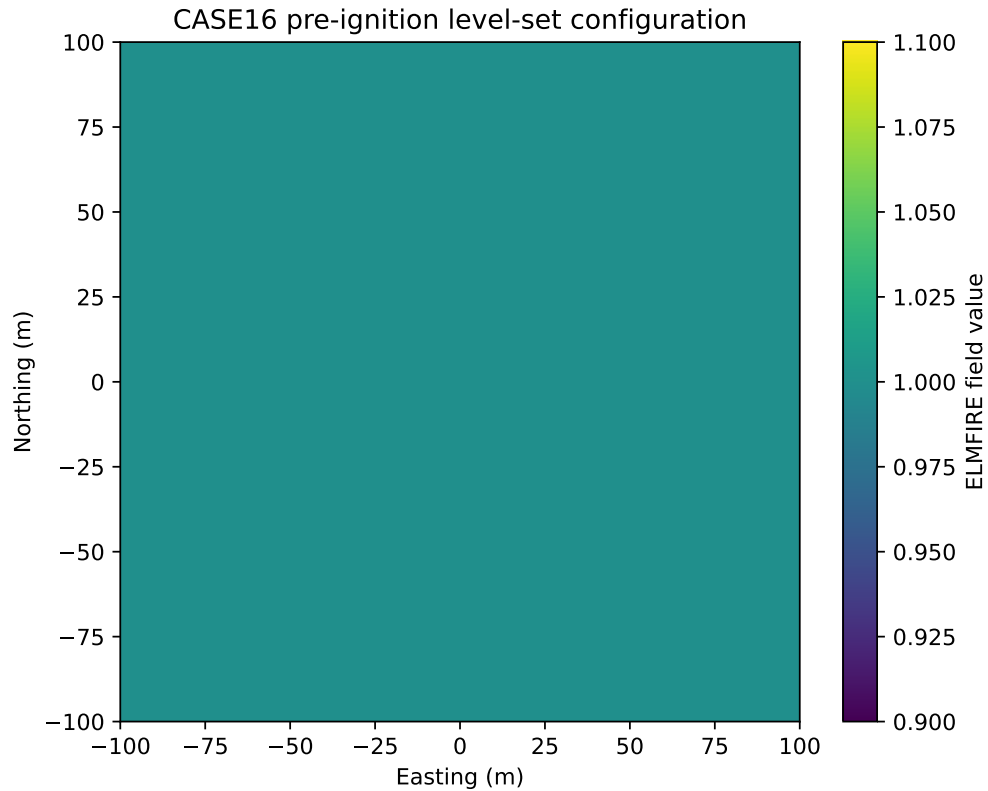


Figure 1: Prepared input conditions from case-local generated rasters. Coordinates are in metres; the colorbar identifies the displayed field and its physical units or category. The ignition panel is derived directly from $\phi \leq 0$.

6 Actual simulation results

Only case-local ELMFIRE rasters are read. Current status is **PASS**; 2 of 2 components pass.

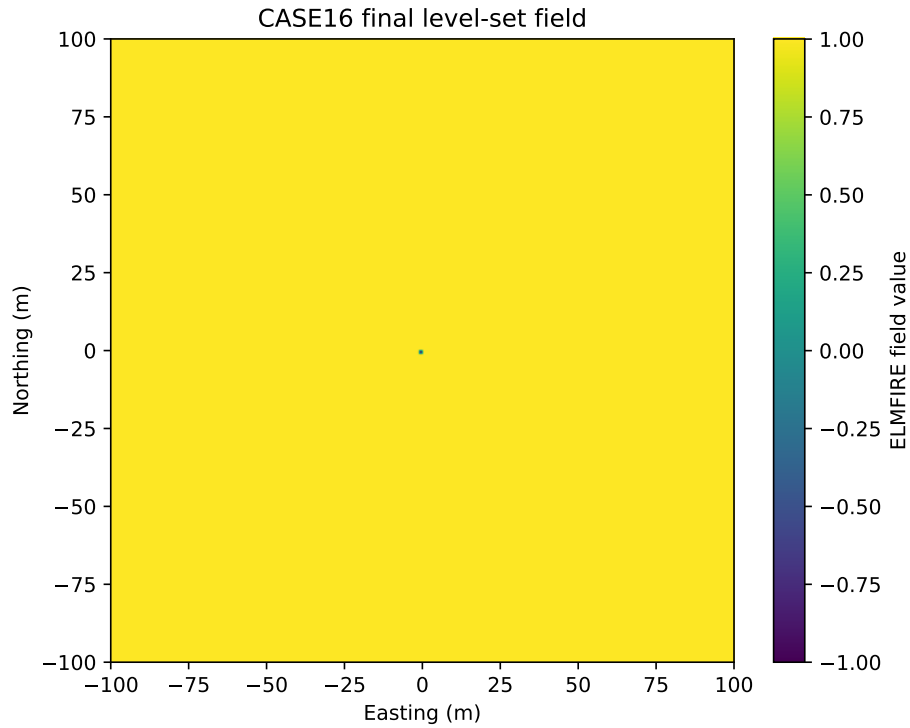


Figure 2: Final whole-domain signed level-set field for the completed zero-spread variant at the final output time. Values are dimensionless; the generated table supplies the rate and confinement limits.

7 Calculated metrics and verification decision

Metric	Expected value or limit	Calculated value	Status
maximum absolute spread rate	0 ($\leq 1\text{e-}06$)	0 m/min	PASS
equivalent burned radius	0 (≤ 2.5 cells (2.5 m))	0.56419 m	PASS

The authoritative overall decision is **PASS**. The generated table is sourced directly from the case-local metrics JSON and is not recomputed in this report.

8 Technical interpretation

A nonzero rate primarily indicates that the zero adjustment was ignored or transformed incorrectly. A confined rate with excessive burned area indicates initialization or level-set evolution behavior. Unavailable output cannot establish the invariant.

9 Scope and limitations

This case verifies a deterministic limiting state and raster adjustment coupling. It does not establish accuracy for positive rates, physical extinction, or any coupled fire process.